DFI101 — BASIC COMPUTER SKILLS FOR DIGITAL FORENSICS

Lab1B

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System Identification and Evidence Layer Mapping

Introduction

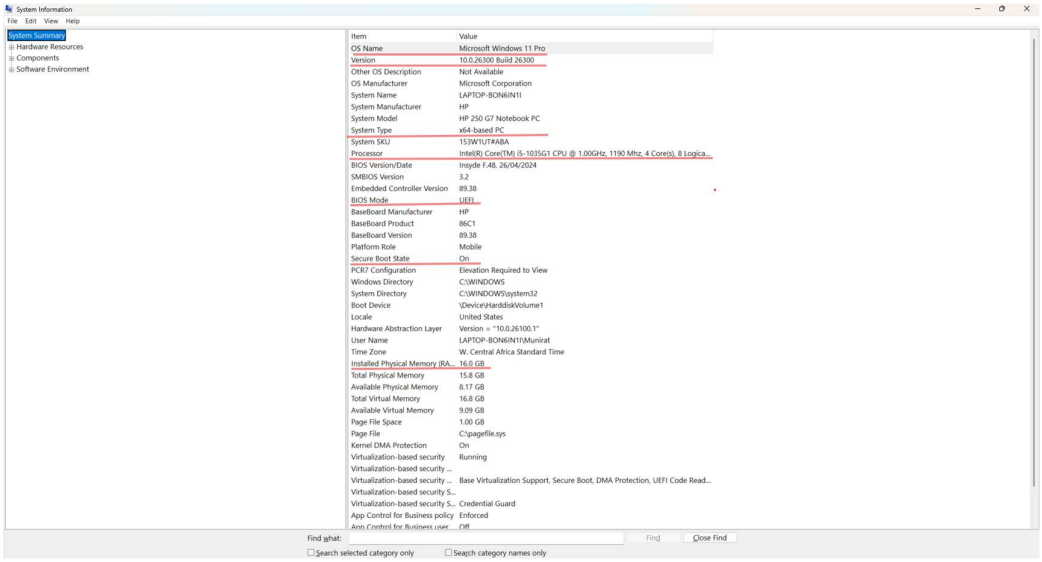
The purpose of this lab is to build a basic understanding of how a computer stores data and starts up, and to apply that knowledge in a digital forensics context. The lab maps evidence sources to system layers, explains the difference between programs, processes, and threads, and shows why process information matters during live triage. It also describes the boot process and identifies whether the system uses BIOS or UEFI. Finally, it verifies the size of selected disk partitions by reading the partition table, recording start sectors and sector counts, and calculating sizes in GB using the sector and CHS formulas. The results are then compared with the sizes reported by the operating system to confirm accuracy and explain any differences.

Tools Used

Category	Tool / Command	Purpose
Operating system	Windows (64-bit, UEFI boot, GPT disk)	Host system used for the practical
Hardware	SK hynix NVMe SSD (476.94 GB)	Disk examined
Command-line tool	Windows PowerShell	Used to run all disk and partition commands
Commands	Get-Disk, Get-Partition, Get-Volume	Shows disk size, partition style (GPT), and sector sizes, partition numbers, types, offsets, and sizes. Shows file system type and OS-reported volume sizes
Windows utility	Disk Management (diskmgmt.msc)	Shows partitions and file systems visually for comparison
Windows utility	System Information (msinfo32)	Confirms BIOS Mode (UEFI) and Secure Boot state
Software	Mobile Calculator	Used to check size calculations
Software	Microsoft Word	Used to prepare the report
Software	Snipping tool	Used for capturing screenshots

A1. Capture basic system specification

System Information: Shows the OS Name as Windows 11 Pro, the BIOS Mode as UEFI – Unified Extensible Firmware Interface, Secure Boot is turned on with RAM of 16GB. The system type x64-based.
Win + R > enter msinfo32 > click on Enter



Command line System Info: This shows the same as the msinfo32.

Launched cmd > enter systeminfo > clicked on Enter, to get the device specs

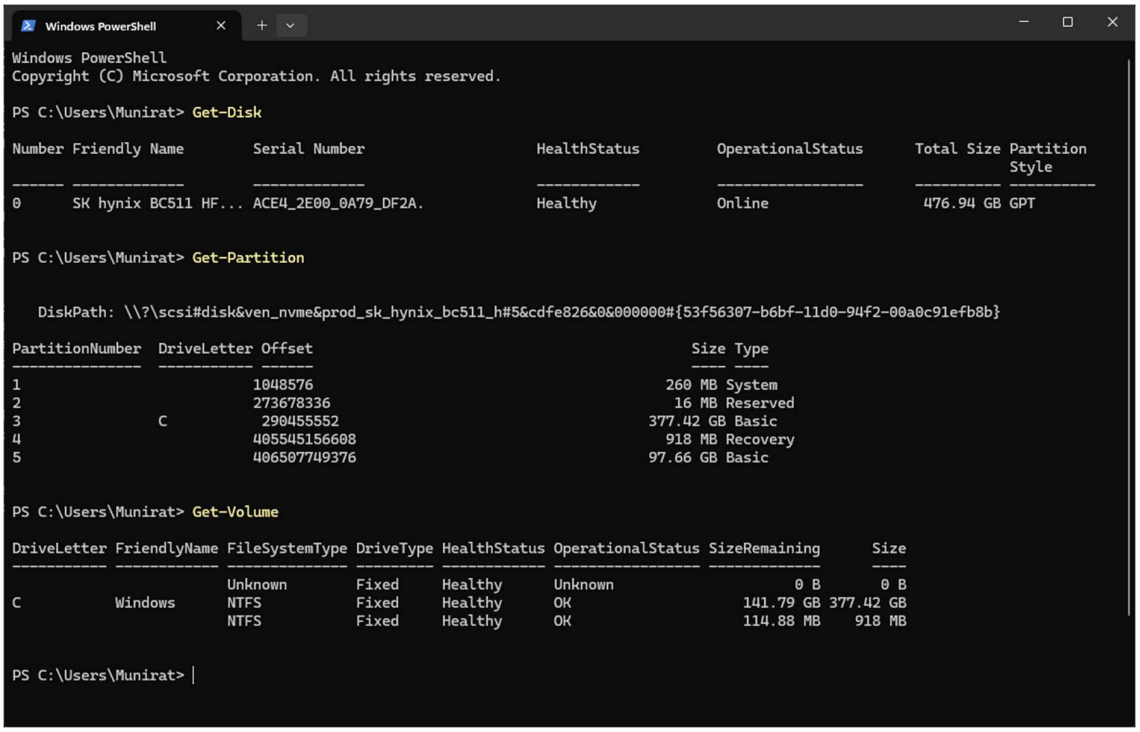
```
Microsoft Windows [Version 10.0.26300.9711]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Munirat>systeminfo

Host Name:                 LAPTOP-BON6IN1I
OS Name:                   Microsoft Windows 11 Pro
OS Version:                10.0.26300 N/A Build 26300
OS Manufacturer:           Microsoft Corporation
OS Configuration:          Standalone Workstation
OS Build Type:             Multiprocessor Free
Registered Owner:          Munirat
Registered Organization:   HP
Product ID:                00330-53359-74051-AAOEM
Original Install Date:     25/10/2024, 05:42:44
System Boot Time:          02/11/2026, 11:54:58
System Manufacturer:       HP
System Model:              HP 250 G7 Notebook PC
System Type:               x64-based PC
Processor(s):              1 Processor(s) Installed.
                           [01]: Intel64 Family 6 Model 126 Stepping 5 GenuineIntel ~991 Mhz
BIOS Version:              Insyde F.48, 26/04/2024
Windows Directory:         C:\WINDOWS
System Directory:          C:\WINDOWS\system32
Boot Device:               \Device\HarddiskVolume1
System Locale:             en-us;English (United States)
Input Locale:              en-us;English (United States)
Time Zone:                 (UTC+01:00) West Central Africa
Total Physical Memory:     16,166 MB
Available Physical Memory: 8,024 MB
Virtual Memory: Max Size:  17,190 MB
Virtual Memory: Available: 8,827 MB
Virtual Memory: In Use:    8,363 MB
Page File Location(s):     C:\pagefile.sys
Domain:                    WORKGROUP
Logon Server:              \\LAPTOP-BON6IN1I
Hotfix(s):                 6 Hotfix(s) Installed.
                           [01]: KB5127694
                           [02]: KB5054156
                           [03]: KB5121794
                           [04]: KB5127648
                           [05]: KB5124807
                           [06]: KB5127647
Network Card(s):           9 NIC(s) Installed.
                           [01]: ExpressVPN TAP Adapter
                                 Connection Name: Ethernet 2
                                 Status:          Media disconnected
                           [02]: Realtek RTL8822CE 802.11ac PCIe Adapter
                                 Connection Name: Wi-Fi
                                 DHCP Enabled:    Yes
                                 DHCP Server:     10.147.37.102
```

A2. Capture storage information

Using PowerShell command, the Get-Disk shows the harddisk size, the partition style and health status. The get-partition shows all the partitions, their sizes and the type.



A3. Map evidence sources to system layers

Guide: User activities = Applications + Operating System + Disk/Memory

Layer / Source	Example Evidence	Where It May Be Found	Why It Matters
Applications Layer	Browser history, cookies, cache, and downloads; email messages and attachments; chat logs; document edit history; app-specific logs	User profile folders (such as AppData, ~/Library), app databases, and cloud-synced app folders	Shows what the user did within the application, such as which sites was visited from browser or file downloaded
Network layer	Firewall and router logs, DNS queries, proxy logs, packet captures, connection records, IP addresses	Routers, firewalls, servers, ISPs, and network monitoring tools	Shows communication between devices, data transfers, and external connections, and can corroborate activity found on the endpoint

Layer / Source	Example Evidence	Where It May Be Found	Why It Matters
OS layer	Event logs, registry keys (e.g.,UserAssist, RecentDocs, USB history), user account and login records	Registry hives, Windows\System32\winevt\Logs,system and user profile directories	Shows system-level activity such as logons, program execution, device connections,
Disk layer	File and folder metadata (created, modified, accessed timestamps), deleted files, file slack, unallocated space, Recycle Bin contents	Hard drive or SSD partitions, file system structures, hidden or unallocated areas	Preserves persistent evidence, including deleted or hidden data, and confirms when files were created, changed, or removed
Memory/process layer	Running processes, open network connections, loaded DLLs, encryption keys, passwords in clear text, injected code, clipboard contents	RAM capture, process list	Captures volatile evidence that is lost at shutdown, which is especially valuable for malware, fileless attacks, and active sessions

Why Evidence May Exist in More Than One Layer

A single user action usually passes through several layers, and each layer records its own trace of it. For example, if a user opens a document from a USB drive and emails it:

- Application layer: The word processor logs it in its recent files list, and the email client stores the sent message and attachment.
- Operating system layer: The registry and Jump Lists record the file access and the USB device connection, and event logs record the session.
- Disk layer: File system timestamps show when the file was accessed or copied, and deleted copies may remain in unallocated space.
- Memory layer: The document content or email session may still be in RAM while the system is running.
- Network layer: Mail server and firewall logs record the outbound transmission.

Evidence appears in multiple layers for several reasons:

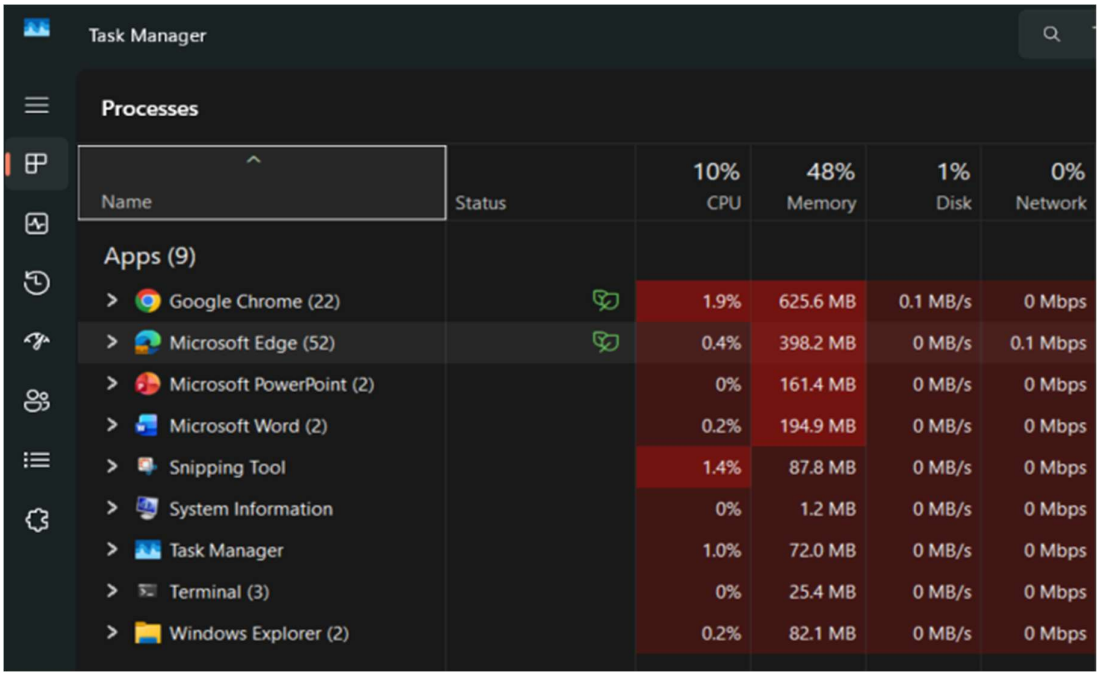
1. Layered design: Applications rely on the operating system, which relies on storage and memory, so activity leaves traces at each level.
2. Different purposes: Each layer records data for its own reasons, such as performance, convenience, auditing, or recovery.
3. Persistence differences: Memory is volatile while disk data persists, so an item deleted from one layer may survive in another.
4. Corroboration: Matching evidence across layers strengthens reliability and helps show that one layer was not altered or tampered with.
5. Anti-forensics resilience: If a user deletes an application's history, traces often remain in the OS artifacts, disk structures, or network logs.

Part B - Program, Process and Thread Investigation

This section links the slide concept of program, process, and thread to forensic observation.

B1. Observe running processes

Right clicked on the task bar > opened the Task Manager and captured the running process below:



Used the PowerShell windows to run get process running:

```
PS C:\Users\Munirat> Get-Process | Select-Object -First 20 Name,Id,CPU,WorkingSet

Name                    Id CPU       WorkingSet
----                    -- ---       ----------
AggregatorHost        9164             12730368
ai                    6364 0.53125     43397120
ai                    6860 0.671875    24145920
ai                   18056 7.375       45539328
AppActions           15972 0.84375     41783296
AppHelperCap          4288             34861056
ApplicationFrameHost  5016 0.421875    52559872
```

B2. Explain the difference between a program, a process, and a thread in your own words.

Program, Process, and Thread

Program: A program is a set of instructions stored on disk, such as chrome.exe or a Word installer. It is passive. It does nothing until someone runs it, much like a recipe sitting in a cookbook.

Process: A process is a program that is currently running. When you open a program, the operating system loads it into memory and gives it its own resources: memory space, a process ID (PID), open files, and security permissions. Using the recipe example, the process is the actual act of cooking. You can run the same program several times, and each run is a separate process.

Thread: A thread is the smallest unit of work inside a process. A process can have one or many threads, and they share the process's memory and resources while each runs its own task. For example, a web browser may use one thread to load a page, another to play a video, and another to respond to your clicks. In the cooking example, threads are the individual cooks working in the same kitchen.

- **Explain why process information may be useful during malware analysis or live forensic triage.**

Process information shows what a system is actually doing at the moment of investigation. Malware must run to cause harm, so it will almost always appear as a process or inside one. Looking at running processes helps investigators in these ways:

1. **Spotting suspicious or unknown processes:** Odd names, misspellings of legitimate names (such as svch0st.exe instead of svchost.exe), or processes running from unusual locations like Temp or AppData can signal malware.
2. **Understanding parent-child relationships:** Knowing which process started another can reveal an attack chain. For example, Word launching PowerShell is unusual and may indicate a malicious macro.
3. **Detecting hidden or injected code:** Some malware hides inside a legitimate process by injecting code into it. Examining loaded DLLs and threads can expose this.
4. **Identifying network activity:** Process information shows which process has open network connections, which can reveal communication with a command-and-control server or data being stolen.
5. **Finding what the process accessed:** Open files, registry keys, and handles show what the process was reading, changing, or creating.
6. **Capturing volatile evidence:** Process details exist only while the system is running and are lost at shutdown. Collecting them early in live triage preserves evidence such as running commands, passwords, or encryption keys held in memory.
7. **Supporting timelines and scoping:** Process start times, user accounts, and command-line arguments help determine when an incident began, who or what triggered it, and which systems may be affected.
8. **Guiding the response:** Triage lets investigators quickly decide whether to isolate the machine, dump memory, or terminate a malicious process, without first performing a full disk analysis.

Part C - Disk Sector and Disk Size Calculation

C2. Calculate your own disk or partition size:

I used the PS cmd Get-Pertition on my device to get my disk details and I noticed PS showed sector as offset so I will be dividing the offset value with 512 bytes;

Start sector = Offset ÷ sector size
Number of sectors = Size in bytes ÷ sector size

```
PS C:\Users\Munirat> Get-Partition


   DiskPath: \\?\scsi#disk&ven_nvme&prod_sk_hynix_bc511_h#5&cdfe826&0&000000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

PartitionNumber  DriveLetter Offset                                        Size Type
---------------  ----------- ------                                        ---- ----
1                            1048576                                    260 MB System
2                            273678336                                   16 MB Reserved
3                C           290455552                                 377.42 GB Basic
4                            405545156608                                918 MB Recovery
5                            406507749376                               97.66 GB Basic
```

for Partition 1:
Start sector: 1,048,576 ÷ 512 = 2,048
Number of sectors: Size = 260 MB, so 260 × 1,048,576 = 272,629,760 bytes, and 272,629,760 ÷ 512 = 532,480 sectors

Partition	**Type**	**Start Sector (Offset ÷ 512)**	**Number of Sectors**	**Sector Size**	**File System**
1	System	2,048	532,480	512 B	Unknown
3	Basic (C:)	567,296	791,513,088	512 B	NFTS
5	Basic	793,960,448	204,806,641	512 B	NTFS

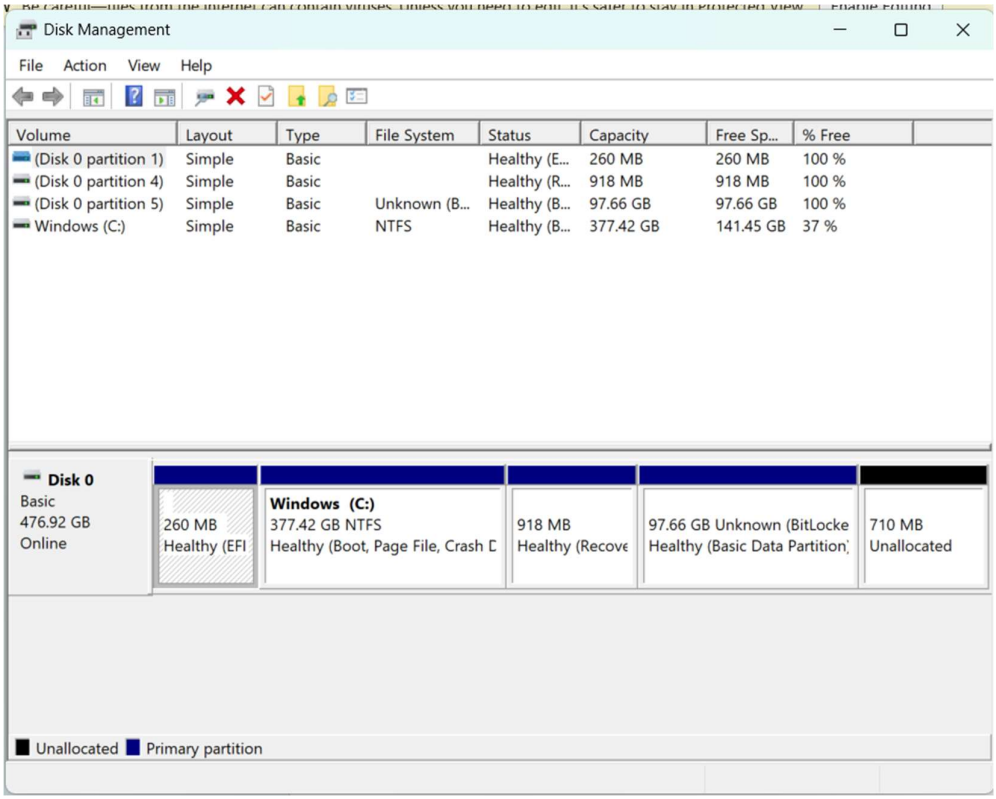
Get-Partition shows partition 5 as 97.66 GB:
97.66 GB × 1024 × 1024 × 1024 ≈ 104,861,000,000 bytes
÷ 512 ≈ 204,800,000 sectors, rounded to a neat number

Size in GB = (sectors × 512) ÷ 1024^3

Partition	Calculation	Size
1	532,480 × 512 = 272,629,760 B÷1024^3	0.25 GB (260 MB)
3	791,513,088 × 512 = 405,254,701,056 B÷1024^3	377.42 GB
5	~204,800,000 × 512 = 104,857,600,000 B÷1024^3	97.66 GB

Part D - Safe Partition Table Inspection

Using Wins+ R key > typed diskmgmt.msc > clicked Enter. Below is the display



Used PS command to get the Disk, Partition and volume as seen below:

```
PS C:\Users\Munirat> Get-Disk

Number Friendly Name Serial Number                  HealthStatus         OperationalStatus      Total Size Partition
                                                                                                         Style
------ ------------- -------------                  ------------         -----------------      ---------- ----------
0      SK hynix B... ACE4_2E00_0A79_DF2A.           Healthy              Online                  476.94 GB GPT


PS C:\Users\Munirat> Get-Partition


   DiskPath: \\?\scsi#disk&ven_nvme&prod_sk_hynix_bc511_h#5&cdfe826&0&000000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

PartitionNumber  DriveLetter Offset                                        Size Type
---------------  ----------- ------                                        ---- ----
1                            1048576                                    260 MB System
2                            273678336                                   16 MB Reserved
3                C           290455552                                 377.42 GB Basic
4                            405545156608                                918 MB Recovery
5                            406507749376                               97.66 GB Basic


PS C:\Users\Munirat> Get-Volume

DriveLetter FriendlyName FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining      Size
----------- ------------ -------------- --------- ------------ ----------------- -------------      ----
                         Unknown        Fixed     Healthy      Unknown                    0 B        0 B
C           Windows      NTFS           Fixed     Healthy      OK                  141.45 GB 377.42 GB
                         NTFS           Fixed     Healthy      OK                  114.88 MB    918 MB


PS C:\Users\Munirat> Get-Disk | Select Number, LogicalSectorSize, PhysicalSectorSize, Size

Number LogicalSectorSize PhysicalSectorSize         Size
------ ----------------- ------------------         ----
     0               512              4096 512110190592
```

Part E - MBR, GPT and Boot Process Explanation

E1. Explain the boot process

Simple flow

Power On → POST → BIOS/UEFI firmware → Read CMOS/firmware settings → Find boot device → Read boot record (MBR or GPT/EFI) → Boot loader/manager → Kernel loads → OS takes control

Explanation:

Power-On

- The user presses the power button, and the power supply delivers electricity to the motherboard, CPU, memory, and other hardware.

POST (Power-On Self-Test)

- The system checks essential hardware components such as RAM, CPU, keyboard, and storage devices to ensure they are functioning correctly.

BIOS/UEFI Starts

- The **BIOS (Basic Input/Output System)** or modern **UEFI (Unified Extensible Firmware Interface)** firmware begins running immediately after POST.

Read CMOS/Firmware Settings

- BIOS/UEFI reads configuration settings stored in **CMOS memory** or firmware storage, including boot order, date/time, and hardware settings.

Find Boot Device

- Using the configured boot order, BIOS/UEFI searches for a bootable device such as an SSD, HDD, USB drive, or network location.

Load MBR/GPT or EFI Boot Records

- On older systems, BIOS loads the **Master Boot Record (MBR)** from the selected drive.
- On modern UEFI systems, the firmware accesses the **GUID Partition Table (GPT)** and loads boot files from the **EFI System Partition (ESP)**.

Start Boot Loader/Boot Manager

- The boot loader (such as **Windows Boot Manager** or **GRUB**) starts and prepares the operating system for loading.

Load Operating System Kernel

- The boot loader loads the operating system kernel into memory, along with essential drivers and system files.

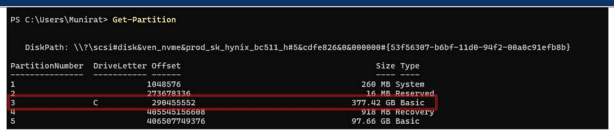
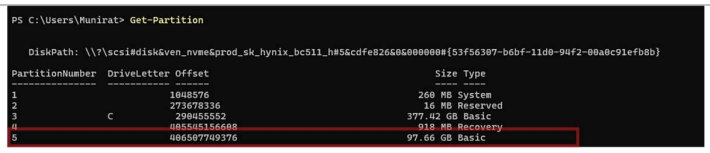
Operating System Takes Control

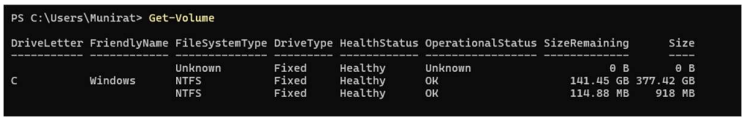
- The kernel initializes hardware and system services, starts user processes, and displays the login screen or desktop. The operating system is now fully in control of the computer.

My PC uses UEFI as seen in the screenshot below:

Item	Value
OS Name	Microsoft Windows 11 Pro
Version	10.0.26300 Build 26300
Other OS Description	Not Available
OS Manufacturer	Microsoft Corporation
System Name	LAPTOP-BON6IN1I
System Manufacturer	HP
System Model	HP 250 G7 Notebook PC
System Type	x64-based PC
System SKU	153W1UT#ABA
Processor	Intel(R) Core(TM) i5-1035G1 CPU @ 1.00GHz, 1190 Mhz, 4 Core(s), 8 Logica...
BIOS Version/Date	Insyde F.48, 26/04/2024
SMBIOS Version	3.2
Embedded Controller Version	89.38
BIOS Mode	UEFI
BaseBoard Manufacturer	HP
BaseBoard Product	86C1
BaseBoard Version	89.38

Part G - Final Practical Assignment

Task	Required Evidence	Student Notes / Answer
Identify partition 3	Screenshot of partition table and command output	 Command ran: Get-partition
Record partition 3 sector count	Start sector, end sector or sector count	Sector size: 512 bytes Start sector = 290,455,552 ÷ 512 = **567,296** Number of sectors = **791,513,088** End sector = 567,296 + 791,513,088 − 1 = **792,080,383**
Calculate partition 1 size	Formula and result in GB	Size = Sectors × Bytes per sector 791,513,088 × 512 = 405,254,701,056 bytes ÷ 1024 = 395,756,544 KB ÷ 1024 = 386,481 MB ÷ 1024 = **377.42 GB**
Identify partition 5	Screenshot of partition table and command output	
Record partition 5 sector count	Start sector, end sector or sector count	Sector size: 512 bytes Start sector = 406,507,749,376 ÷ 512 = **793,960,448** Number of sectors = **204,800,000** End sector = 793,960,448 + 204,800,000 − 1 = **998,760,447**

Task	Required Evidence	Student Notes / Answer
Calculate partition 5 size	Formula and result in GB	204,800,000 × 512 = 104,857,600,000 bytes ÷ 1024 = 102,400,000 KB ÷ 1024 = 100,000 MB ÷ 1024 = **97.66 GB**
Compare with OS-reported size	Disk Management, Get-Volume, lsblk, or df output	For Partition 3, Get-Volume shows C: Size = 377.42 and Disk Management shows about 377.42 GB. For Partition 5, Get-Volume reports 97.66 GB and Disk Management shows about 377.42 GB as well 
Explain difference if any	Short explanation of rounding, reserved space, file system overhead, or GiB/GB differences	Any tiny difference (a few KB) comes from rounding to two decimals